

Assisting with Voluntary Pinch Force Control by Using Electrical Muscle Stimulation and Active Bio-Acoustic Sensing

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ABSTRACT

We present a novel wearable system for assisting with voluntary pinch force control that requires no devices on the fingers. We use active bio-acoustic sensing to estimate voluntary pinch force with piezo elements attached to the back of the hand and electrical muscle stimulation to the forearm to control involuntary pinch force in a closed-loop system. We conducted a user study with eight participants to investigate whether the system could assist with pinch force control under target forces of 3 N, 6 N, or 9 N. When they tried to get pinch force closer to each target force with our system, the medians of the absolute errors were 0.8 N, 1.0 N, and 2.8 N, respectively. When they tried to do it without our system, they were 0.9 N, 1.4 N, and 2.4 N, respectively.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

ems, active bio-acoustic sensing, closed-loop control

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1 INTRODUCTION

Electrical Muscle Stimulation (EMS) can steer the user's limbs; thus, previous studies have proposed ways of helping users achieve good form and posture in playing sports and instruments by using EMS [7, 14]. Fingertip force control is also important for motor skills learning; sensitive force control is required for athletes in various sports such as golf [8] and tennis [2]. Our study aims to explore the usage of EMS to support fingertip force control. As a first step, we focus on lateral pinch force, which is grip force achieved with the pad of the thumb and the lateral surface of the index finger to grasp objects [15]. There are two forearm muscles

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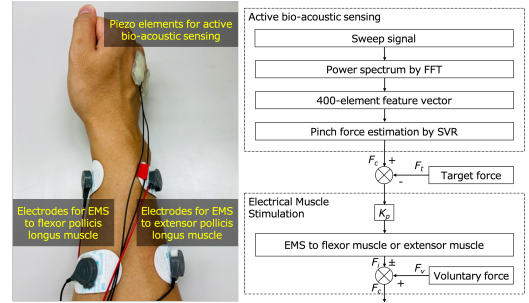


Figure 1: Overview of our system. Here, F_c is current pinch force, F_t is target force, F_v is voluntary pinch force, F_i is involuntary pinch force, and K_p is proportional gain.

that are mainly relevant to the pinch force: the flexor pollicis longus muscle and the extensor pollicis longus muscle. Thus, the target muscles that we have to consider are only the two muscles; this fact is suited to the first step.

For assisting with controlling pinch force in a closed-loop system, it is necessary to measure the force with wearable or embedded sensors. For example, it is possible to measure fingertip force with sensors worn on the fingers [10]. Exoskeleton gloves [13] can also assist with fingertip force control. However, devices attached to the fingers would interfere with the user's finger manipulation and change the tactile sensation. By using electromyogram (EMG) sensors on the forearm, it is possible to estimate fingertip force with no devices on the fingers [1, 6]. However, the current of EMS would affect measured signals with EMG sensors. Therefore, it is needed to consider how to reduce the noises when EMS and EMG sensors are used at the same time [5, 11].

We propose a novel wearable system that assists with pinch force control by using EMS and active bio-acoustic sensing [9]. Active bio-acoustic sensing is based on the principle of active acoustic sensing [12], and it can be used for micro-gesture recognition and discrimination of different levels of touch-force. We use a sensing method for estimating pinch force that uses a machine learning regression model. The advantage of combining active bio-acoustic sensing and EMS is that the user's fingers will be free and possible to use active bio-acoustic sensing while presenting EMS because the current of EMS does not interfere with acoustic signals.

2 PROPOSED METHOD

We use active bio-acoustic sensing and EMS in a closed-loop system. An overview of our system is shown in Figure 1.

2.1 Active Bio-Acoustic Sensing

The system consists of two piezo elements (THRIVE, K2512BP1) attached to acrylic plates with hot glue. The piezo elements are connected to an audio interface (Steinberg, UR22mkII). The piezo elements are placed on the back of the hand; the locations are 10 mm from the metacarpal joints as based on the work of Kubo et al. [9].

The system emits sinusoidal sweep signals from 20–40 kHz increasing linearly in intervals of 20 ms repeatably. The sampling rate is 96 kHz. Audio signals are converted to the power spectrum by using a fast Fourier transform (FFT) with 8192 samples. From the power spectrum of 20–40 kHz, it constructs a 400-element feature vector using a peak detection algorithm to obtain valid values for the force estimation engine. Using the features as input, we use support vector regression (SVR) for the force estimation. The parameters are set on the basis of grid search; “kernel” is “rbf”, “gamma” is 1, and “C” is 10.

2.2 Electrical Muscle Stimulation

We generate and use EMS by using a function generator (Nihon Kohden Corporation, SEN-8203) and isolators (Nihon Kohden Corporation, SS104-J) connected to pairs of disposable electrodes (Nihon Kohden Corporation, F-150M). The electrodes are placed on the flexor pollicis longus muscle and the extensor pollicis longus muscle of the right forearm. A microcomputer (Arduino, Arduino Uno) is connected to the function generator to emit a trigger.

For involuntary pinch force control, we use proportional control to change the intensity of EMS. We use pulse frequency modulation (PFM) control to do so; the higher the frequency is, the higher the intensity is [4]. The microcomputer connected to the function generator is used for PFM control. The frequency of EMS to the flexor pollicis longus muscle and that to the extensor pollicis longus muscle are controlled as follows.

$$f_f = \begin{cases} 200 & (K_p(F_t - F_c) \geq 200) \\ K_p(F_t - F_c) & (K_p(F_t - F_c) < 200) \\ 0 & (K_p(F_t - F_c) \leq 0) \end{cases} \quad f_e = \begin{cases} 200 & (K_p(F_c - F_t) \geq 200) \\ K_p(F_c - F_t) & (K_p(F_c - F_t) < 200) \\ 0 & (K_p(F_c - F_t) \leq 0) \end{cases}$$

Here, f_f is the frequency of EMS to the flexor pollicis longus muscle, f_e is the frequency of EMS to the extensor pollicis longus muscle, K_p is the proportional gain, F_t is the target force, and F_c is the current force. We set K_p to 20 on the basis of tuning.

3 USER STUDY

We evaluated the effectiveness of our system in assisting with pinch force control. The experiment was approved by our institution’s ethical review board. We recruited eight healthy participants (two female, 20s–40s) from our local institution. All participants signed a consent form before starting the experiment.

3.1 Procedure

The participants pinched a force sensor (IMADA, ZTS-200N). They were asked to keep the target force for 10 seconds five times in 10-second intervals (total of 100 seconds for each trial). There were three target forces: 3 N, 6 N, and 9 N. The current of EMS to the flexor pollicis longus muscle was set at 7–8 mA, and that to the extensor pollicis longus muscle was set at 10–11 mA. Each participant conducted the following tasks.

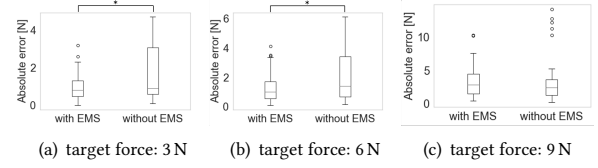


Figure 2: Absolute error distribution with/without EMS for all participants under each condition.

- (1) There was a training phase. One target force was selected randomly. Participants looked at the monitor of the force sensor and pinched the force sensor while controlling the force from 0–12 N for about 100 seconds to collect training data of 1000 samples. During this task, EMS was applied depending on the difference of the target force and the current force measured by the force sensor.
- (2) There was a test phase. We covered the monitor of the force sensor so that the participants could not look at it. Then, they tried to keep the target force for 10 seconds five times in 10-second intervals with or without EMS. These conditions were selected randomly. Under the EMS condition, EMS was applied depending on the difference in the target force and current force estimated by the SVR model.
- (3) There was another test phase. Participants tried to keep the target force for 10 seconds five times in 10-second intervals without or with EMS [opposite condition of (2)].

3.2 Results

We calculated the medians of the force sensor value for each task. The number of pieces of data for each condition (e.g., a target force of 3 N, with EMS) was 40 (5 times $\times$ 8 participants). Figure 2 shows the distribution of the absolute error under each condition. When they tried to get the pinch force closer to each target force with EMS, the medians of the absolute errors were 0.80 N, 1.0 N, and 2.8 N, respectively. When they tried to do so without EMS, they were 0.90 N, 1.4 N, and 2.4 N, respectively. We conducted a Wilcoxon signed-rank test with Bonferroni correction to investigate whether there was a significant difference between them; the results showed that there were significant differences under the target forces of 3 N ($Z = -3.0, p = 0.002 < 0.05/3, r = 0.47$) and 6 N ($Z = -2.6, p = 0.007 < 0.05/3, r = 0.41$), and there was not a significant difference under the target force of 9 N ($Z = 0.32, p = 0.75, r = 0.05$). The effect sizes were medium effects under the target forces of 3 N and 6 N, and negligible effects under the target forces of 9 N [3]. The means of the absolute errors of force estimation while keeping the pinch force were 1.76 N, 1.67 N, and 3.36 N, respectively.

The system gave incorrect feedback to the users when the force estimation accuracy was low. The effectiveness of EMS assistance depends on the accuracy of the force estimation; the assistance will be more effective when the accuracy of the force estimation increases. We consider that the boundary condition of piezo elements was changed during the tasks due to skin deformation when the target force was 9 N. We will investigate the best position of piezo elements to prevent boundary condition changes.

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